

py1.py U X py2.py U py3.py U py4.py U py5.py U

CO2-AT1 > py1.py > ...

```
1 words = ["connected", "connecting", "connection"]
2 print("-" * 70)
3 print("{:<15} {:<12} {:<10} {:<15} {:<15}".format(
4     "Word", "Root", "Suffix", "Type", "Normalized"))
5 print("-" * 70)
6 for word in words:
7     if word.endswith("ed"):
8         root = word[:-2]
9         suffix = "ed"
10        t = "Inflectional"
11    elif word.endswith("ing"):
12        root = word[:-3]
13        suffix = "ing"
14        t = "Inflectional"
15    elif word.endswith("ion"):
16        root = "connect"
17        suffix = "ion"
18        t = "Derivational"
19    print("{:<15} {:<12} {:<10} {:<15} {:<15}".format(
20        word, root, suffix, t, "connect"))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS> python py1.py
no 2] No such file or directory
(.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS> cd CO2-AT1
(.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> python py1.py
-----
Word          Root      Suffix    Type      Normalized
-----
connected     connect  ed        Inflectional  connect
connecting    connect  ing       Inflectional  connect
connection    connect  ion       Derivational  connect
(.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1>
```

py1.py U py2.py U X py3.py U py4.py U py5.py U

CO2-AT1 > py2.py > ...
1 words = ["unhappy", "happiness", "happily"]
2 print("-" * 85)
3 print("{:<12} {:<10} {:<10} {:<10} {:<15} {:<10}".format(
4 "Word", "Prefix", "Root", "Suffix", "Type", "Normalized"))
5 print("-" * 85)
6 for word in words:
7 prefix = "-"
8 suffix = "-"
9 if word.startswith("un"):
10 prefix = "un"
11 root = "happy"
12 elif word.endswith("ness"):
13 suffix = "ness"
14 root = "happy"
15 elif word.endswith("ly"):
16 suffix = "ly"
17 root = "happy"
18 print("{:<12} {:<10} {:<10} {:<10} {:<15} {:<10}".format(
19 word, prefix, root, suffix, "Derivational", "happy"))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> python py1.py  
connection      connect      ion      Derivational      connect  
● (.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> python py2.py  
-----  
Word      Prefix      Root      Suffix      Type      Normalized  
-----  
unhappy      un      happy      -      Derivational      happy  
happiness      -      happy      ness      Derivational      happy  
happily      -      happy      ly      Derivational      happy  
❖ (.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1>
```

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py1.py U py2.py U **py3.py U X** py4.py U py5.py U

CO2-AT1 > py3.py > ...
1 words = ["played", "player", "playing"]
2 print("-" * 75)
3 print("{:<12} {:<10} {:<15} {:<15} {:<10}".format(
4 "Word", "Stem", "Removed Affix", "Type", "Normalized"))
5 print("-" * 75)
6 for word in words:
7 if word.endswith("ed"):
8 stem = word[:-2]
9 affix = "ed"
10 t = "Inflectional"
11 elif word.endswith("ing"):
12 stem = word[:-3]
13 affix = "ing"
14 t = "Inflectional"
15 elif word.endswith("er"):
16 stem = word[:-2]
17 affix = "er"
18 t = "Derivational"
19 print("{:<12} {:<10} {:<15} {:<15} {:<10}".format(
20 word, stem, affix, t, "play"))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> python py2.py  
happily                happy        ly                Derivational        happy  
• (.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> python py3.py  
-----  
Word                Stem                Removed Affix        Type                Normalized  
-----  
played              play                ed                    Inflectional        play  
player              play                er                    Derivational        play  
playing             play                ing                   Inflectional        play  
(.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> 
```

py1.py U py2.py U py3.py U py4.py U X py5.py U

CO2-AT1 > py4.py > ...

```
1 words = ["writes", "writing", "written"]
2 print("-" * 90)
3 print("{:<12} {:<20} {:<10} {:<15} {:<10}".format(
4     "Word", "State Path", "Root", "Type", "Normalized"))
5 print("-" * 90)
6 for word in words:
7     if word == "writes":
8         path = "q0 -> q1 -> q2"
9         root = "write"
10        t = "Regular"
11    elif word == "writing":
12        path = "q0 -> q1 -> q3"
13        root = "write"
14        t = "Regular"
15    elif word == "written":
16        path = "q0 -> q4"
17        root = "write"
18        t = "Irregular"
19    print("{:<12} {:<20} {:<10} {:<15} {:<10}".format(
20        word, path, root, t, "write"))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

(.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> python py3.py

playing play ing Inflectional play

● (.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> python py4.py

```
-----
Word      State Path      Root    Type      Normalized
-----
writes    q0 -> q1 -> q2    write   Regular   write
writing   q0 -> q1 -> q3    write   Regular   write
written   q0 -> q4          write   Irregular  write
```

❖ (.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> |

py1.py U py2.py U py3.py U py4.py U py5.py U X

CO2-AT1 > py5.py > ...
1 from nltk.stem import PorterStemmer
2 ps = PorterStemmer()
3 words = ["relational", "relation", "relate"]
4 print("-" * 80)
5 print("{:<15} {:<25} {:<20} {:<15}".format(
6 "Word", "Rule Applied", "Intermediate", "Final Stem"))
7 print("-" * 80)
8 for word in words:
9 if word == "relational":
10 rule = 'Removed "ional"'
11 intermediate = "relate"
12 elif word == "relation":
13 rule = 'Removed "ion"'
14 intermediate = "relate"
15 elif word == "relate":
16 rule = 'Removed "e"'
17 intermediate = "relat"
18 stem = ps.stem(word)
19 print("{:<15} {:<25} {:<20} {:<15}".format(
20 word, rule, intermediate, stem))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> python py4.py  
written      q0 -> q4      write      Irregular      write  
● (.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1> python py5.py  
-----  
Word          Rule Applied          Intermediate          Final Stem  
-----  
relational    Removed "ional"        relate               relat  
relation      Removed "ion"          relate               relat  
relate        Removed "e"            relat               relat  
❖ (.venv) PS C:\Users\bandi\OneDrive\Documents\NLP - PROGRAMS\CO2-AT1>
```